ADVERTISING SALES PREDICTION *USING POLYNOMIAL REGRESSION*

Introduction

- Advertising is one of the most important factors that influence product sales.
- Companies spend money on different advertising channels such as:
  - TV
  - Radio
  - Newspaper
- Understanding how advertising affects sales helps businesses make better decisions.
- This project uses **advertising sales data** to analyze and predict sales.
- Machine Learning helps analyze and predict sales more accurately

Objective

- To study the relationship between advertising spending and product sales.
- To identify which advertising channel has the strongest impact on sales.
- To predict product sales using machine learning
- To improve prediction using Polynomial Regression
- To support better business decision-making.

Problem Statement

- Businesses invest in multiple advertising platforms.
- They need to know:
  - Which platform contributes most to sales
  - How much sales can be expected from a given budget
- Manual estimation is difficult and may not be accurate.
- A predictive model can help estimate future sales based on ad spending.

Project Overview

- This project predicts **Sales** using advertising expenditure data.
- Dataset used: **advertising.csv**
- Features:
  - **TV**
  - **Radio**
  - **Newspaper**
- Target:
  - **Sales**
- Model used:
  - **Polynomial Regression**
- Goal:
  - Improve prediction accuracy by capturing **non-linear relationships**.

Why Polynomial Regression?

- Linear Regression assumes a straight-line relationship
- Real-world advertising data may contain:
  - Curved patterns
  - Interaction effects
  - Non-linear growth
- Polynomial Regression helps by adding:
  - squared terms
  - interaction terms

Formula Idea

Polynomial Regression = PolynomialFeatures + LinearRegression

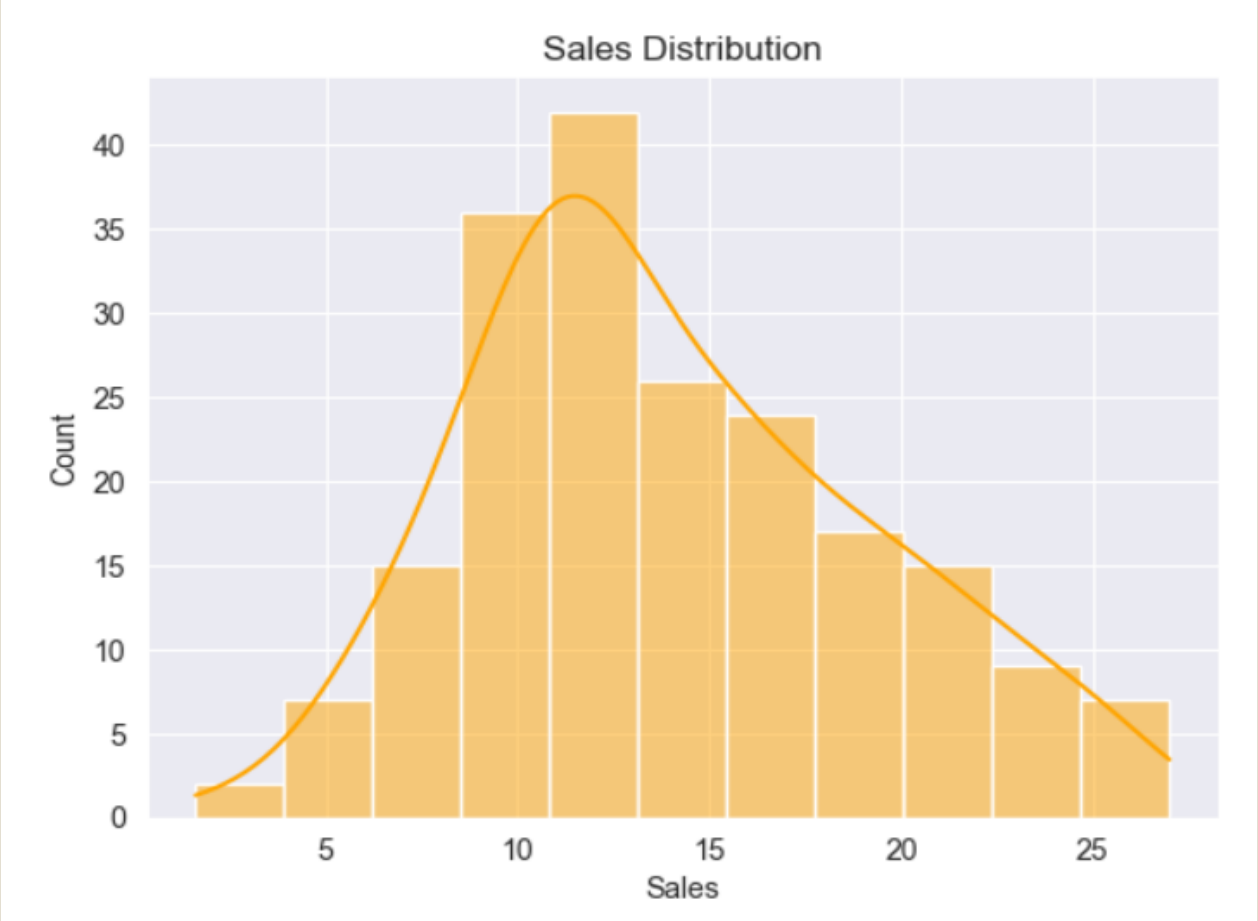
Data Preprocessing

- Loaded advertising.csv using pandas
- Checked for null values and duplicates
- Verified data types of all columns
- Separated input features (X) and target variable (y)
- Performed train-test split for evaluation

Exploratory Data Analysis (EDA)

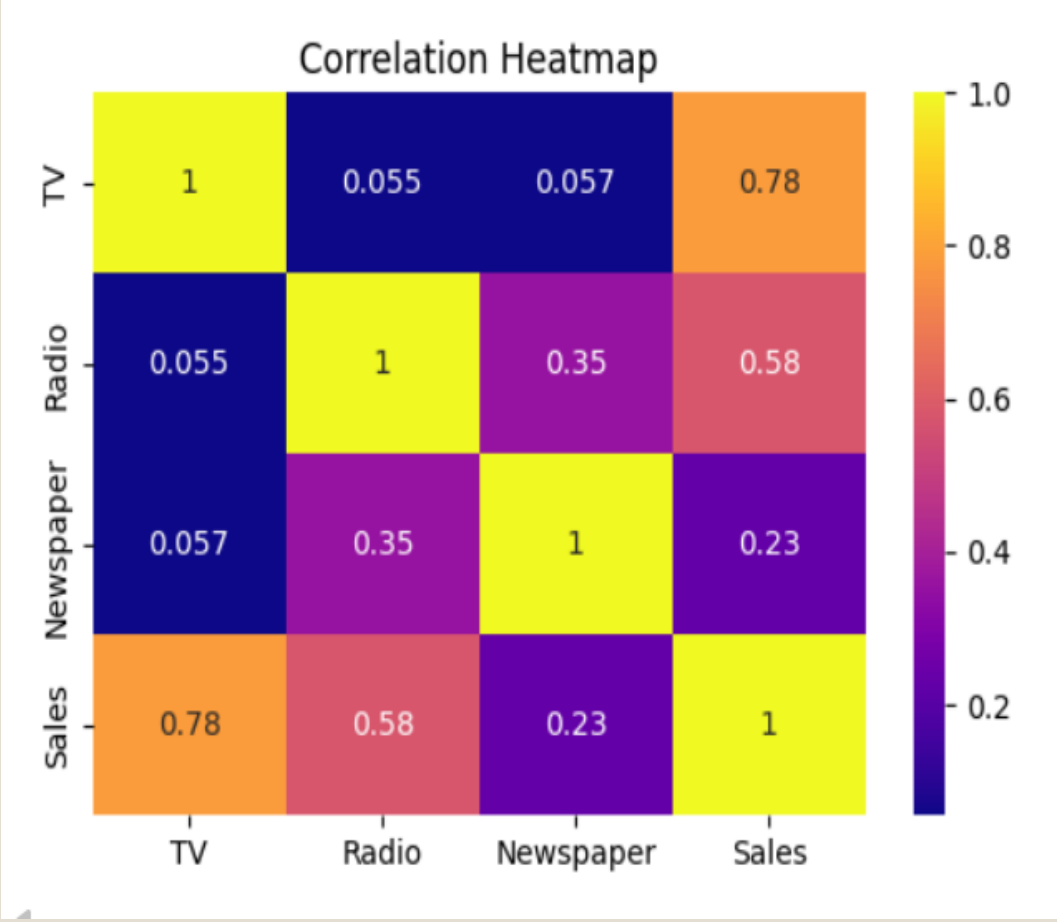
Visualizations Used

- **Pair plot**
- **Correlation heatmap**
- **Histograms with KDE**
- **Scatter Plots**
- **Regression Plots**



Key Insights from EDA

- **TV** has the strongest positive relationship with Sales
- **Radio** also shows a significant impact
- **Newspaper** has weaker correlation
- Sales increase as TV and Radio spending increase
- The data is suitable for regression analysis



Model Building

- Split data into training and testing sets
- Apply Polynomial Feature Transformation
- Train **LinearRegression()** on transformed data
- Predict on test data
- Compare predictions with actual sales

Model Used

Polynomial Regression (Degree = 2)

Model Evaluation Metrics

- **R^2 Score**
  - Measures how much variance in Sales is explained
- **Mean Squared Error (MSE)**
  - Measures average squared prediction error
- **RMSE**
  - Square root of MSE for easier interpretation

Goal

- Higher **R^2**
- Lower **MSE**

Report Results

- R^2 Score (Accuracy): 0.9869
- Mean Squared Error (MSE): 0.4129
- Interpretation: The model explains ~98.69% of the variation in Sales
- Conclusion: Excellent fit for this dataset

Business Insights

- TV advertising has the strongest impact on sales
- Radio advertising also contributes significantly
- Newspaper advertising has relatively smaller influence
- Marketing teams should prioritize high-performing channels
- The model can guide budget optimization decisions

Advantages of This Model

- Simple and easy to interpret
- Fast to train and deploy Works well for linear relationships
- Provides measurable impact of each feature
- Useful for business presentations and decision-making

Conclusion

- Advertising sales data helps understand the effect of ad spending on sales.
- TV and Radio are the most influential advertising channels.
- Linear Regression gives excellent results for this dataset.
- The project demonstrates how machine learning can support business decisions.
- It is a practical and useful project for marketing analytics.

THANK YOU